**6** $p^3 \times p^x = p^9$

(a) Find the value of $x$.

$x =$ .....

**(1)**

$(7^2)^y = 7^{10}$

(b) Find the value of $y$.

$y =$ .....

**(1)**

$100^a \times 1000^b$ can be written in the form $10^w$

(c) Show that $w = 2a + 3b$

**(2)**